JIR_2019.

Temp. u KELVINIMA!

(11.) Idealni plin

$R = 287\ J/kgK$

$k = 1,4$

$\boxed{\Delta c, \Delta z = 0}$

$p_1 = 7\,bar$ $\quad p_2 = 1\,bar$

$\vartheta_1 = 600°C$ $\quad \vartheta_2 = 300°C$

$p_{ok} = 1\,bar$

$\vartheta_{ok} = 15°C$

$q_{12} = -8\,kJ/kg$

a) $W_{t12} = ?$

b) $W_{gub} = T_{ok} \cdot \Delta S_{uk}$

c) $\Delta S_{uk} = ?$

c_p:

$k = \frac{c_p}{c_v}$

$c_p = c_v + R$

$c_v = \frac{c_p}{k}$

$c_p\left(1 - \frac{1}{k}\right) = R$

$\underline{c_p = 1004,5\ J/kgK}$

a) OTV. SUST.:

$q_{12} + h_1 = W_{t12} + h_2$

$W_{t12} = q_{12} + h_1 - h_2$

$W_{t12} = q_{12} + c_p(T_1 - T_2)$

$\boxed{W_{t12} = 293,35\,kJ/kg}$

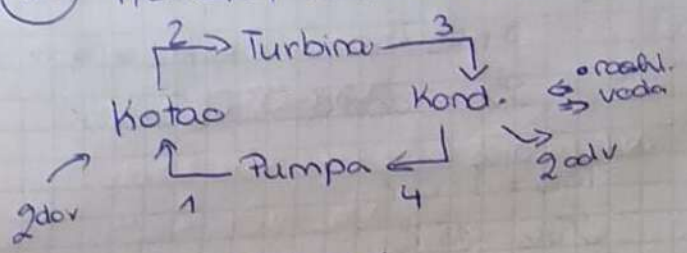
b) $W_{gub} = T_{ok} \cdot \Delta S_{uk} = \underline{47,08\,kJ/kg}$

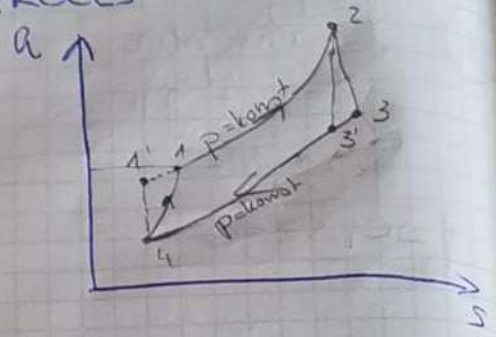
c) $\Delta S_{uk} = \Delta S_{pl} + \Delta S_{ok} = 163,385\ J/kgK$

$\Delta S_{pl} = \Delta S_s = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} = 135,622\ J/kgK$

$\Delta S_{ok} = \frac{|q|}{T_{ok}} = 27,763\ J/kgK$

(12.) REALNI RANKINEOV KR. PROCES





$P_t = 1000\,MW$

$p_1 = p_2 = 8{,}5\,MPa$

$\vartheta_2 = 650\,°C$

$p_3 = p_4 = 10\,kPa$

$\eta_t = 0{,}9$

$\eta_p = 0{,}85$

10kPa: $h' = h_4 = 191{,}8\,kJ/kg$

$h'' = h_3 = 2585\,kJ/kg$

$v' = 0.001\,m^3/kg$

$h = h_2 = 3756\,kJ/kg$

$s_2 = s_{3'} = 7{,}121\,kJ/kg$

10kPa: $h = h_3' = 2256\,kJ/kg$

a) $\eta = ?$ (uzeti u obzir rad pumpanja!)

b) $x = ?$

c) $\dot{m}_{pare,turb.} = ?$

b) $x = \dfrac{h - h'}{h'' - h'} = \boxed{0{,}863}$

turb. $\eta_t = \dfrac{W_{realt}}{W_{idt}}$ pumpa $\eta_p = \dfrac{W_{idp}}{W_{realp}}$

a) $\eta = \dfrac{W_{realt.} + W_{realp.}}{q_{dov}}$

$\boxed{\eta = 0{,}332}$

$\eta_t = \dfrac{W_{23'}}{W_{23'}} = \dfrac{h_2 - h_3}{h_2 - h_3'}$

$W_{realt.} = h_2 - h_3 = 1171\,kJ/kg$

$q_{dov} = q_{12} = h_2 - h_1 = 3554{,}212\,kJ/kg$

$W_{idp} = W_{41'} = h_{1'} - h_4 = v'(p_1 - p_4)$

$W_{idp} = 8{,}490\,kJ/kg$

c) $\dot{m} \Rightarrow$

$P_t = \dot{m} W_t$

$\dot{m} = \dfrac{P_t}{W_{realt}} = \boxed{853{,}97\,kg/s}$

$W_{realp} = \dfrac{W_{idp}}{\eta} = 9{,}988\,kJ/kg$

$h_1 = 9{,}01\,kJ/kg$

(13.) (1.) PWR NE

$n = 3$

$P_j = 3\,GW$

$m_u = 65\,t = m_{UO_2} \cdot \frac{238}{270}$

$\Phi = 3 \cdot 10^{17}\ n/m^2 s$

$\sigma_f = 580 \cdot 10^{-28}\ m^2$

chladiac: $\dot{m}_{cl} = 5000\ kg/s$

$c_p = 5{,}7\ kJ/kg$

$\rho = 720\ kg/m^3$

$\vartheta_{ul} = 295°C$

$h_{FW} = 391\ kJ/kg$

$h_{ST} = 2764\ kJ/kg$

$\dot{m}_{pare} = 1269\ kg/s$

a) $e = ?$

b) $\vartheta_{izl} = ?$

c) $P_p = ?$

d) $\Delta P_p = ?$

a) e:

$N_{U-235} = e\ m_u \frac{N_A}{235}$

$P_j = 200 \cdot 1{,}6 \cdot 10^{-13} \cdot N\ \sigma_f \cdot \Phi$

$N = 5{,}388 \cdot 10^{28}$

$\rightarrow e = 0{,}0323$

$\boxed{e = 3{,}23\%}$

b)

$P_j = \dot{m}_{chlad} \cdot c_p \cdot \Delta T \cdot n$

$\Delta T = 35.087\ K / °C$

$\Delta T = \vartheta_{izl} - \vartheta_{ul}$

$\vartheta_{ul} = 330{,}087°C = 603{,}24\,K$

c)

$P_T = \dot{m}_{pare} \cdot \Delta h = P_j + n \cdot P_p$

$\Rightarrow P_p = 3{,}78\ MW$

d) $P_p = \dot{m}_{chlad} \cdot v \cdot \Delta P_p$

$\Delta P_p = \frac{P_p \cdot \rho}{\dot{m}_{chlad}} = 544{,}32\,kPa$

(14.) Tlačni tunel

(2.) $H_{zah} = 200\,m$ n.v.

$H_t = 50\,m$ n.v.

$H_{GV} = 40\,m + H_{zah} = 240\,m$ n.v.

$H_{DV} = 45\,m$ n.v.

$Q_i = 150\,m^3/s$

$\eta = 0{,}85$

$D = 3\,m$

a) ASPIRATOR

$$H_w = H_{GV} - H_t - \frac{c^2}{2g} = \underbrace{189{,}99\,m}\approx 190\,m$$

$Q = c \cdot A$

$c = \frac{A}{Q} = 0{,}047\,m/s$ $\quad \underbrace{P = 237{,}63\,MW}$

$A = \frac{D^2\pi}{4} = 7{,}069\,m^2$

b) DIFUZOR

$D = 5\,m$ $\quad c = 0{,}131\,m/s$

$A = 19{,}635\,m^2$

$$H_n = H_{GV} - H_{DV} - \frac{c^2}{2g} = \underbrace{194{,}99\,m}\approx 195\,m$$

$\underbrace{P = 243{,}90\,MW}$

c) $m = 0{,}7$ $\quad m = \frac{W}{8760 \cdot P}$

$W_a = 8760 \cdot m \cdot P = \underbrace{1{,}457\,TWh}$

$W_b = \underbrace{1{,}496\,TWh}$

(16.) (4.) VA $P_N = 1{,}5\,MW \Rightarrow v_N = 11\,m/s \Rightarrow 20\%$ god.

$P = 0{,}7\,MW \Rightarrow v = 8\,m/s \Rightarrow 40\%$ god

$\rho = 1{,}225\,kg/m^3$ $\eta = 1$

a) $W_{el,god} = ?$

b) $m = ?$

c) $D = ?$, $c_{pe} = 0{,}4 \Rightarrow v_n$

a) $W = 8760 \cdot \eta \cdot c_{pe}\, 0{,}5 \cdot \rho \cdot A \cdot (0{,}2 \cdot v_N^3 + 0{,}4 \cdot v_8^3)$

$W_{god} = 4{,}65\,GWe$

c) $D = ?$

$P = \eta \cdot c_p \cdot 0{,}5\rho \cdot A \cdot v^3$

$A = 4599{,}89\,m^2 = \frac{D^2 \pi}{4}$

$\Rightarrow D = 76{,}53\,m$

b) $m = \frac{W_{god}}{8760 \cdot P_N} = 0{,}354$